Element	*Mass No.*	*Atomic Mass (AMU)*	*Element*	*Mass No.*	*Atomic Mass (AMU)*
	152	151.919756		187[a]	186.955833
	154	153.922282	$_{76}$Os	184	183.952750
$_{63}$Eu	151	150.919838		186	185.953870
	153	152.921242		187	186.955832
$_{64}$Gd	152	151.919794		188	187.956081
	154	153.920929		189	188.958300
	155	154.922664		190	189.958630
	156	155.922175		192	191.961450
	157	156.924025	$_{77}$Ir	191	190.960640
	158	157.924178		193	192.963012
	160	159.927115	$_{78}$Pt	190[a]	189.959950
$_{65}$Tb	159	158.925351		192	191.961150
$_{66}$Dy	156	155.923930		194	193.962725
	158	157.924449		195	194.964813
	196	195.964967		205	204.974442
	198	197.967895	$_{82}$Pb	204	203.973044
$_{79}$Au	197	196.966541		206	205.974468
$_{80}$Hg	196	195.965820		207	206.975903
	198	197.966756		208	207.976650
	199	198.968279	$_{83}$Bi	209	208.981082
	200	199.968327	$_{90}$Th	232[a]	232.038124
	201	200.970308	$_{92}$U	234[a]	234.040904
	202	201.970642		235[a]	235.043915
	204	203.973495		238[a]	238.050770
$_{81}$Tl	203	202.972353			

[a] Unstable nuclide.

Source: From Weidner, R., and Sells, R. *Elementary Modern Physics*, 2nd edition. Boston, Allyn & Bacon, 1968. Used by permission.

ANSWERS TO SELECTED PROBLEMS

Chapter 2

2-1. 8; 16; 0.1369 amu; 127.5 MeV; 8.0 MeV/nucleon
2-2. 15.999 amu
2-4. Tungsten, 58, 220 eV; hydrogen, 10.1 eV
2-5. 0.51 MeV
2-6. 2.37×10^{24}, 0.85 g
2-7. 15.2×10^{15} J; 3.63×10^{12} kcal

Chapter 3

3-1. 4.6 hr; 6.9 hr; 156 hr
3-2. 3.5×10^{-7}; 6.6×10^{15}; 1.8×10^{-6}
3-4. 36.7%
3-6. 2.4×10^{19} sec^{-1}; 0.124 Å
3-8. 3.14 hr
3-10. 6.15×10^{14}; 9.2×10^{-8} g
3-11. 30.3 mCi
3-12. 1.09 days; 4100 Ci; 16.4 Ci
3-13. ^{74}As, positron decay and/or electron capture; ^{76}As, negatron decay
3-14. 80

Chapter 4

4-1. 2.031 keV/cm
4-2. 51,699 IP/cm
4-4. 0.59
4-5. 0.94 cm
4-7. 12 keV; 35 keV
4-9. 138 keV; 12 keV; decreased
4-10. 865 keV

Chapter 5

5-2. 3.12×10^{17}; 5000 J/sec
5-4. 11.5 degrees
5-5. No; yes
5-6. No; 1 should be eliminated
5-7. 3 per min
5-8. 250 KeV; 0.023; 0.05 A
5-9. 2 mm
5-10. 9.6 mm

Chapter 6

6-1. 1.06×10^9 MeV/(m^2-sec); 1.06×10^{10} meV/m^2
6-2. 0.68 J/kg
6-3. 363 J/m^2; 2.27×10^{15} photons/m^2
6-4. 2.4×10^{11}
6-5. 7.4
6-6. 1.53×10^{-10} A
6-7. 52 picofarads
6-8. 270 R
6-11. 10^{-4} J/g; 10^{-3} J
6-12. 0.85
6-13. 1,500 mrem
6-14. 0.014°C
6-15. 1.44×10^{18}
6-17. (a) 1.5 mm A1; (b) 2.25 mm A1; (c) 0.7

Chapter 8

8-1. (a) No; (b) yes, yes; (c) no
8-2. No
8-4. 4.25×10^6
8-5. 0.6 mV

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Chapter 9

9-1. 3.6×10^{10}
9-2. 10,500; 2000
9-3. 1.60; 60%
9-6. (a) 75 mCi/mL; (b) 0.064
9-9. 2.76 MeV = photopeak; 2.25 MeV = single-escape peak; 1.74 MeV = double-escape peak; 1.38 MeV = photopeak; 0.87 MeV = single-escape peak; 0.51 MeV = annihilation peak; 0.20 MeV = backscatter peak
9-11. About 580
9-14. 220 MCi
9-15. 3400 yr
9-16. 2.84×10^5 Ci/g

Chapter 10

10-1. 5120 bits = 5×2^{10}
10-2. 503,316,480 bits = $8 \times 60 \times 2^{20}$
62,914,560 bytes = 60×2^{20}
10-3. 16,777,216 bits = $32 \times 0.5 \times 2^{20}$
2,097,152 bytes = $4 \times 0.5 \times 2^{20}$
10-4. 20,480 images
10-5. 31.25 mV
10-6. PROM
10-7. 1.8 (line pairs)/mm < 2.5 (line pairs)/mm, does not meet the standard
10-8. 4.8 Mbit

Chapter 11

11-1. a. 6 cpm, 1.67%
b. 4.7 cpm, 3.60%
c. 8 cpm, 3.32%
11-2. a. The precision is described by the 2-mm standard deviation. Accuracy relates the researcher's data to the "true" answer determined by some independent method, which in this case is the study of excised bone. The degree of accuracy is therefore given by the 2% difference between the means of the two methods of bone measurement. The results are biased because the original measurements are consistently larger than the true value.
b. Both mean and standard deviation are "scaled" by the same proportion (see Table 11-3). Corrected mean = (0.98)(5.5) = 5.4 cm. Corrected standard deviation = (0.98)(2 mm) = 1.96 mm = 0.02 cm.
Note: The corrected values are given with the same number of significant figures as the original data, and the same units (cm) are used for both.
11-3. t value = 2.31; $P = 0.022$

Chapter 12

12-1. 59%
12-2. 5500 mL
12-4. 43%
12-5. (a) ^{241}Am; (b) ^{197}Hg

Chapter 13

13-1. 10% and 3%
13-2. 126.5 mR

Chapter 14

14-1. 3240
14-3. Yes
14-4. 9 MHz
14-6. 1.8×10^{-3} c/(kg-min), 7.0 roentgen/min
7.1×10^{-3} c/(kg-min), 27.7 roentgen/min

Chapter 16

16-1. Unsharpness, contrast, noise, distortion and artifacts
16-2. Geometric, subject, motion, receptor, unsharpness
16-3. 0.4 mm
16-4. 1.8 mm
16-5. 24 cm
16-6. Subject, technique, contrast agents, receptor
16-7. Structure noise, radiation noise, receptor noise, quantum mottle

Chapter 17

17-1. 0.2 mm; 2.5 line pairs/mm
17-2. Higher; projected focal spot is smaller
17-3. 0.41

Chapter 18

18-2. The individual sees at 20 ft what an ordinary person sees at 100 ft.
18-4. Sensitivity $= p(\mathrm{TP}) = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$
Specificity $= p(\mathrm{TN}) = \mathrm{TN}/(\mathrm{TN} + \mathrm{FP})$
18-6. Specificity, because a false-negative means that a tumor that should have been detected is missed.

Chapter 19

19-2. -13 dB
19-3. (a) $\alpha_R = 0.011$ $\alpha_T = 0.989$
(b) $\alpha_R = 0.011$ $\alpha_T = 0.989$
19-4. 33.4 degrees
19-6. 6.9 dB

Chapter 20

20-2. 1.2 mm
20-4. 48 cm

Chapter 21

21-2. 128
21-3. $^1/_4$
21-5. Attenuation

Chapter 22

22-1. 325 Hz; higher
22-2. 1.3 kHz
22-5. 650 Hz; a, decrease; b, increase

Chapter 23

23-2. 7.63×10^{18}
23-3. 1.98 tesla
23-4. 1/16 of the maximum signal

Chapter 24

24-1. 11.52 min
24-2. a
24-3. 280 Hz

Chapter 28

28-1. 30 rem
28-2. 300 mrem
28-3. 440 mR
28-4. 2.0 mm Pb
28-5. None
28-6. 0.4 mm Pb
28-7. 0.7 mm Pb
28-8. 1.27 mm Pb

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